BFS:

```python
def water_jug_bfs():
    visited = []
    queue = [((0, 0), [])]   # simple list as queue

    while len(queue) > 0:
        (x, y), path = queue.pop(0)   # dequeue (FIFO)

        if (x, y) in visited:
            continue
        visited.append((x, y))
        path = path + [(x, y)]
        # Goal condition
        if x == 2:
            return path
        # Possible moves
        next_states = [
            (4, y),   # Fill 4L
            (x, 3),   # Fill 3L
            (0, y),   # Empty 4L
            (x, 0),   # Empty 3L
            (x - min(x, 3 - y), y + min(x, 3 - y)),   # 4 → 3
            (x + min(y, 4 - x), y - min(y, 4 - x))    # 3 → 4
        ]

        for state in next_states:
            if state not in visited:
                queue.append((state, path))  # enqueue
    return None
print("BFS Path:")
print(water_jug_bfs())
```

OUTPUT:

```
Executing task: C:\python\python.exe 'd:\5th Semester data\AI\Assignment 1\bfs.py'

BFS Path:
[(0, 0), (4, 0), (1, 3), (1, 0), (0, 1), (4, 1), (2, 3)]
Terminal will be reused by tasks, press any key to close it.
```

DFS

```
def water_jug_dfs():
    visited = []
    stack = [((0, 0), [])]   # simple list as stack

    while len(stack) > 0:
        (x, y), path = stack.pop()   # pop (LIFO)

        if (x, y) in visited:
            continue

        visited.append((x, y))
        path = path + [(x, y)]
        # Goal condition
        if x == 2:
            return path

        # Possible moves
        next_states = [
            (4, y),
            (x, 3),
            (0, y),
            (x, 0),
            (x - min(x, 3 - y), y + min(x, 3 - y)),
            (x + min(y, 4 - x), y - min(y, 4 - x))
        ]

        for state in next_states:
            if state not in visited:
                stack.append((state, path))  # push
    return None
print("DFS Path:")
print(water_jug_dfs())
```

OUTPUT:

```
PS D:\5th Semester data\AI\Assignment 1> & C:/python/python.exe "d:/5th Semester data/AI/As
py"
DFS Path:
[(0, 0), (0, 3), (3, 0), (3, 3), (4, 2), (4, 0), (1, 3), (1, 0), (0, 1), (4, 1), (2, 3)]
PS D:\5th Semester data\AI\Assignment 1>
```

END,,,